

LightGBM Bidirectional Feature Selection Algorithm for Time Series Prediction

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Abstract—In order to improve the prediction accuracy of high-dimensional complex time series, a LGB-BS Feature selection algorithm based on LightGBM is proposed. The algorithm first uses LightGBM algorithm to build an iterative lifting Tree model for the original feature set, and measures the importance of features. Then, two feature importance metrics are used to combine two-way selection strategy to eliminate irrelevant and redundant features and select the optimal feature subset. Finally, the LSTM neural network is used to predict the original feature set and the optimal feature subset, respectively. The experimental results show that compared with the original feature set, the LGB-BS algorithm can effectively improve the prediction accuracy of high-dimensional complex time series by selecting the optimal feature subset, and improve the problem of poor feature input in time series prediction models.

Keywords—Time series prediction, LightGBM, Feature selection, LSTM, Composite model

I. INTRODUCTION

Time series prediction is widely used in various fields of society^[1-3]. Accurate time series prediction is of great significance for social and economic development^[4]. For complex data with multiple features, reasonable and effective feature selection is helpful to improve the accuracy of time series prediction model^[5].

Literature [6] used principal component analysis to extract features from data, but this method only retains a specific percentage of principal components, so some data will be lost. In [7], a feature selection method based on Pearson correlation coefficient is proposed for high-dimensional complex data. This method can not measure the nonlinear relationship and is sensitive to outliers. Reference [8] proposed a feature selection algorithm based on improved genetic algorithm, but it did not overcome the shortcomings of poor local search ability of genetic algorithm. Reference [9] used LightGBM^[10] to perform one-way feature selection on environmental factors based on a single feature importance metric, but the feature subset obtained in this way may be locally optimal.

In summary, based on LightGBM, this paper proposes a new feature selection algorithm LGB-BS (Light Gradient Boosting Machine Bidirectional Search) to solve the problem of insufficient prediction accuracy caused by high-dimensional complex data in the time series prediction process.

II. LGB-BS-LSTM MODEL AND LGB-BS ALGORITHM

A. LGB-BS-LSTM model

The LGB-BS-LSTM model established in this paper is divided into three modules, which are data preprocessing module, LGB-BS feature selection module and LSTM

prediction module. The LGB-BS-LSTM model structure is shown in Figure 1.

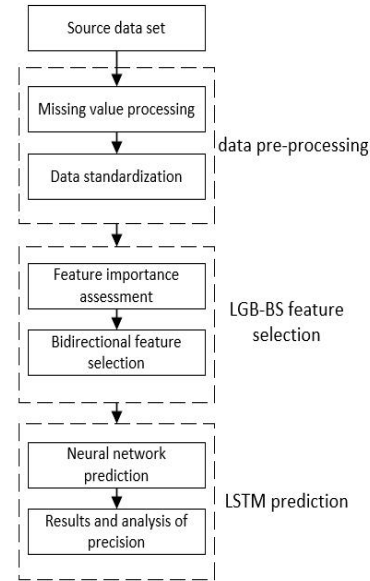


Figure.1 LGB-BS module structure

The data preprocessing module completes the missing values in the original input data by KNN interpolation method, and performs data standard operation on the input data, so that different attributes in the data can be in the same order of magnitude after data standardization, which helps the model to better fit the data.

The LGB-BS feature selection module performs feature selection on the processed data, eliminates irrelevant features and redundant features, and selects the optimal feature subset.

The LSTM prediction module inputs the selected optimal feature subset into the LSTM neural network training prediction, and analyzes the prediction results.

B LGB-BS algorithm

The detailed structure of the LGB-BS feature selection module is shown in Figure 2.

Step1 : Enter the data, establish the root node and build the decision tree model with the root node as the starting point, and set the maximum tree model depth.

Step2 : Traverse all the features $\{X_1, \dots, X_n\}$ in the data X_i and calculate the gain of each feature. In LightGBM, the evaluation criterion for selecting a segmentation point for segmentation is Gain. Gain calculation formula is :

$$Gain = \frac{1}{2} \left[\frac{(\sum_{i \in I_L} g_i)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{(\sum_{i \in I_R} g_i)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda} \right]$$

Through calculation, the feature with the largest gain is selected as the segmentation point for segmentation.

Step3 : Traverse the segmentation points in all decision

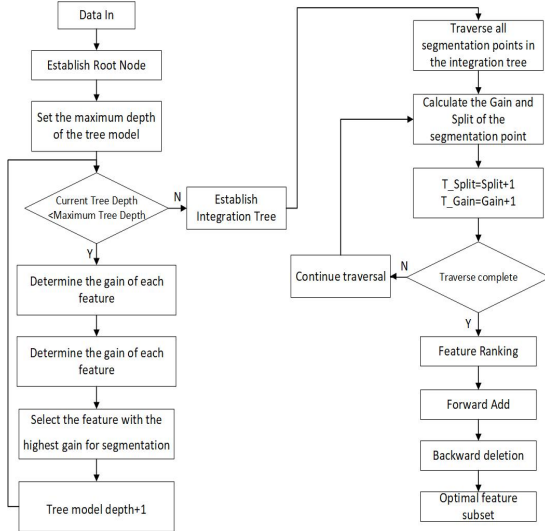


Figure.2 The detailed structure of LGB-BS module

The LGB-BS algorithm obtains the candidate feature set A by sorting the features according to the Gain feature importance evaluation index through the fast sorting algorithm. LGB-BS uses a bidirectional search strategy to extract feature subset from candidate feature set A. The bidirectional search strategy is divided into two parts, which are the forward addition process and the backward deletion process. The specific process is shown in Figure 3 and Figure 4 :

Add the specific process forward :

Step1 : Enter the feature set A sorted by the feature importance index Gain and Create a blank set M. Select the feature with the highest importance in the set A to add to the set M in turn.

Step2 : If the performance of the evaluation function is improved after adding the feature, the feature is retained, otherwise, the feature is discarded.

Step3 : Perform the above steps many times until there is no feature in the set A that can join the set M, and then the target feature subset M is obtained.

The specific process of backward deletion :

Step1 : The feature subset M selected by the forward addition process is sorted from large to small according to the feature importance index Split. Remove the features with the lowest importance in the set M in turn.

Step2 : If the performance of the evaluation function is improved after the feature is deleted, the feature is deleted, otherwise the feature is retained.

trees in the integrated tree, and pass the formula $T_Gain = \sum_{t=1}^N Gain_t$, $T_Split = \sum_{t=1}^N Split_t$. The total gain brought by the features in the ensemble tree and the total number of segmentations are calculated respectively, and the features are sorted according to the Gain feature importance index.

Step4 : The optimal feature subset is selected by filtering the features through forward addition and backward deletion.

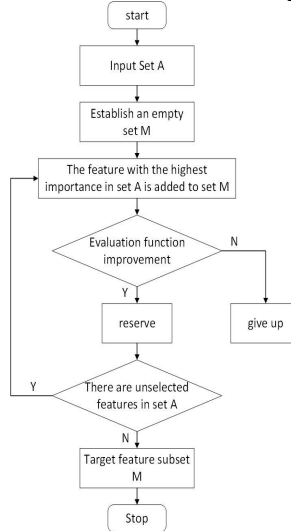


Figure.3 Forward Add Process

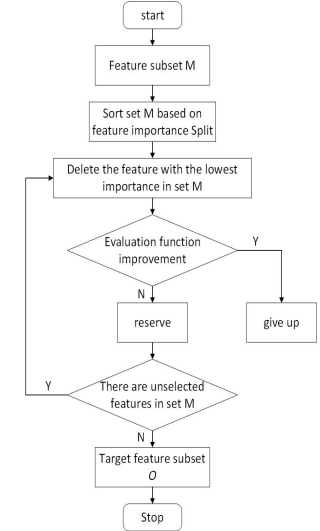


Figure.4 Backward deletion process

Step3 : Perform the above operations multiple times until there is no feature in the set M that can be deleted. At this time, the set M is the final target feature subset O.

Compared with the traditional feature selection algorithm, the advantage of the LGB-BS feature selection algorithm proposed in this paper is to select two feature importance metrics Split and Gain as the basis for cross-floating bidirectional feature selection. To a certain extent, it reduces the limitations brought by a single feature importance metric and optimizes the direction of feature selection.

III. EXPERIMENT

A. Experimental data

In this paper, three different time series data sets are used for empirical analysis : SML2010 data set, Beijing multi-site air quality data set and London shared bicycle data set. SML2010 data set and Beijing multi-site air quality data set are from UCI open database of University of California, and London shared bicycle data set is from iDataScience platform.

This paper uses three different time series data sets : SML2010 data set, Beijing multi-site air quality data set and London shared bicycle data set for empirical analysis. This paper includes three experiments. Experiment 1 predicts the indoor temperature at time t through the SML2010 data set ; experiment 2 predicts the PM2.5 concentration at the Temple of Heaven site at time t ; experiment 3 predicts the rental of shared bicycles in London at time t.

B. Evaluation Metric

This paper will use three indicators : RMSE, MAE, MAPE to evaluate the prediction effect. The smaller the values of

RMSE, MAE and MAPE are, the more accurate the model prediction is. The specific formula is as follows :

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}$$

$$MAE = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right|$$

C Experimental environment

The computer environment used in the experiment is as follows : the CPU is Intel core i5-11400H, the operating system is a 64-bit win10 system, and the software used is pycharm, python3.9 version, LSTM, LightGBM, etc. All are carried out under the PyTorch framework.

D. Experimental analysis

(1) Data preprocessing : In the three data sets used in the experiment, there are missing values in the data set of the Temple of Heaven site. Firstly, the missing values are supplemented, and then the data set is standardized.

(2) Single model prediction : LSTM model is established to predict three data sets.

(3) Construction of LGB-BS-LSTM model : The LGB-BS algorithm is used to select the optimal feature subset from the data set, and the feature subset is input into the LSTM model for prediction.

(4) Model analysis : MSA, RMSE, MAPE three evaluation indicators LSTM model and LGB-BS-LSTM model were used for analysis and comparison.

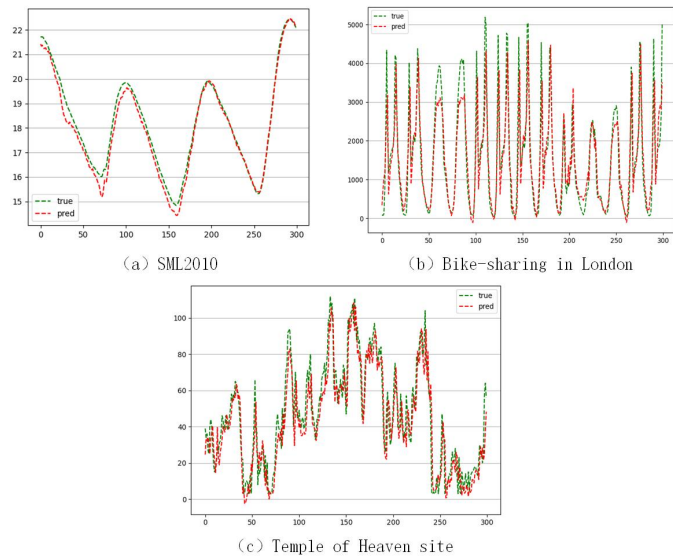
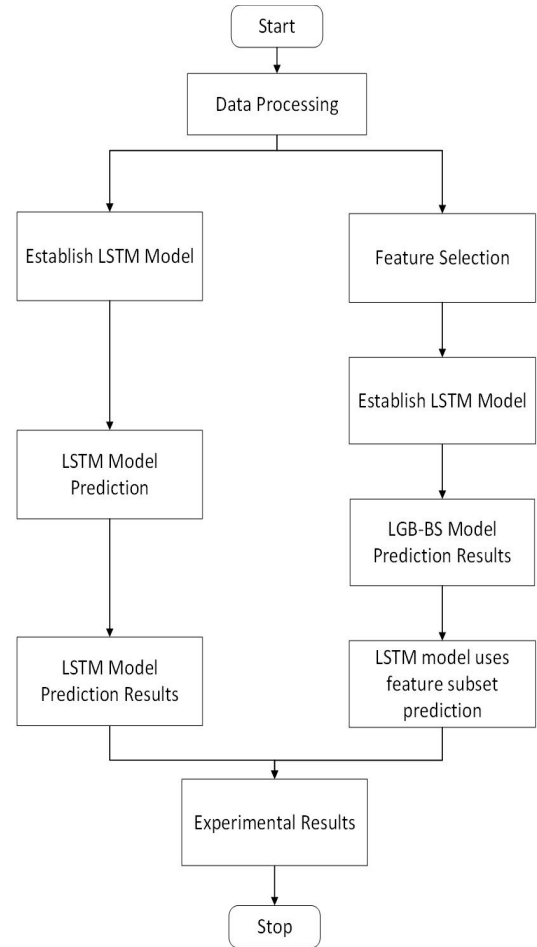


Figure.6 LSTM model prediction results

The green curve true is the true value, and the red curve

The experimental process is shown in Figure 5



Through experiments, the prediction results of LSTM model and LGB-BS-LSTM model for three data are obtained respectively. As shown in Figs.5 and 6.

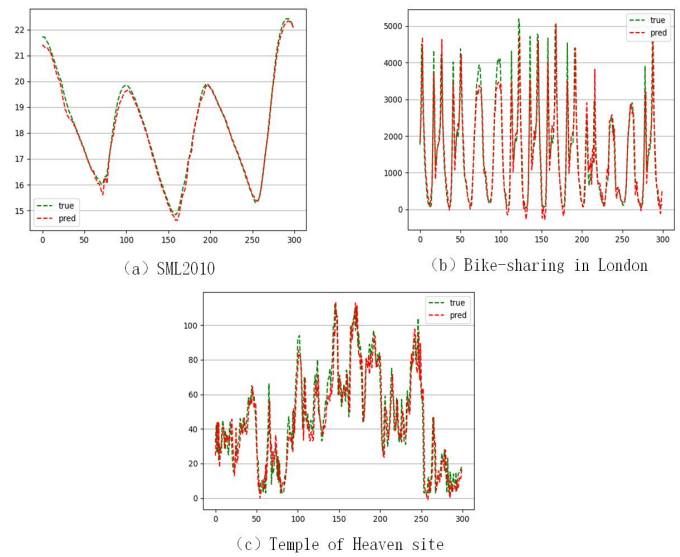


Figure.7 LGB-BS model prediction results

pred is the predicted value of the LGB-BS-LSTM model.

Comparing the prediction results of the two models, it can be seen that the prediction results of the LGB-BS-LSTM model are more accurate than the LSTM model, and the prediction curve is closer to the real curve.

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be seen that the prediction results of the LGB-BS-LSTM model are more accurate than the LSTM model, and the prediction curve is closer to the real curve.

The three evaluation indexes of the two models are as follows:

Table 1 Experimental evaluation index

Model	Experiment 1			Experiment 2			Experiment 3		
	RMSE	MAE	MAP	MA	RMS	MAP	MAE	RMSE	MAP
LSTM	0.1645	0.1212	E	E	E	E	167.72	109.22	0.291
			1	62	3				
LGB-BS	0.1302	0.1055	0.005	17.1	9.883	0.297	162.09	104.28	0.253
-LSTM			5	21			0	8	

Compared with the first experiment, it can be seen that the RMSE, MAE and MAPE values of the LGB-BS-LST model are 0.0343,0.0157 and 0.0005 lower than those of the LSTM model.

Through experiment 2, it can be seen that the RMSE, MAE and MAPE values of the LGB-BS-LSTM model are 0.641,0.44 and 0.024 lower than those of the LSTM model.

Experiment 3 shows that the RMSE, MAE and MAPE values of the LGB-BS-LSTM model are 5.634,4.932 and 0.038 lower than those of the LSTM model.After comparing the above three experimental results, it is obvious that the prediction accuracy of the LGB-BS-LSTM model is higher than that of the LSTM model. Secondly, by comparing the LSTM model and the LGB-BS-LSTM model, it can be found that the LGB-BS-LSTM model has a greater improvement in prediction accuracy than the LSTM model without feature selection. Through the comparison between Experiment 2 and Experiment 1 and Experiment 3, it can be seen that the LGB-BS-LSTM model has no obvious improvement in the prediction effect of PM2.5 concentration compared with the other two experiments. Because the indoor temperature in the SML2010 data set has a strong correlation with light, outdoor temperature and relative humidity, and the bicycle rental in the London shared bicycle data set has a strong correlation with holidays, real temperature and somatosensory temperature, while the concentration of PM2.5 is not strongly correlated with dew point temperature, wind speed and wind direction. Although the prediction of PM2.5 concentration by the LGB-BS-LSTM model in Experiment 2 is not significantly improved compared with the LSTM model, the prediction accuracy of the LGB-BS-LSTM model is significantly improved compared with the LSTM model in Experiment 1 and Experiment 3.

After comparing the above three experimental results, it is obvious that the prediction accuracy of the LGB-BS-LSTM model is higher than that of the LSTM model. Secondly, by comparing the LSTM model and the LGB-BS-LSTM model, it can be found that the LGB-BS-LSTM model has a greater improvement in prediction accuracy than the LSTM model without feature selection. Through the above experiments, we can see that for high-dimensional complex data, the accuracy of time series prediction is significantly improved after using

LGB-BS algorithm to select features from the data set.

IV. CONCLUSIONS

In this paper, a bidirectional feature selection algorithm LGB-BS based on LightGBM is proposed. The algorithm uses a bidirectional search strategy combined with the two importance indicators of LightGBM to select the features of the data. The experimental results show that the LGB-BS algorithm can effectively extract the optimal feature subset in the data and change the problem of poor feature input in the time series prediction process. Therefore, for high-latitude complex data with a large amount of data and a large number of features, feature selection can effectively reduce data complexity and improve the prediction accuracy of the model.

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